MABEL Self-Collision Avoidance: A Toggleable Stop-at-Margin Velocity Damper over the Wrist-Tracking IK

MABEL Simulation Project*

Abstract

MABEL's "Hold" teleoperation mode tracks the operator's wrists with a differential inverse-kinematics (IK) step; unconstrained, the operator can drive a palm straight into the robot's own chest, head, or opposite arm. This document specifies the self-collision-avoidance layer that prevents such commands while preserving tracking everywhere the body is clear. The layer is *runtime-toggleable* and *off by default*, so plain teleop is untouched unless the operator enables it. We wrap the moving links (forearms, hands) and the parts they strike (torso, head) in *capsule* bounding volumes with closed-form signed distance, and derive the distance kinematics $\dot{d} = n^\top (J_a - J_b)\dot{q}$ from per-witness-point Jacobians. The **default "stop" law** keeps the operator's intended *direction* of motion exactly: it uniformly scales the tracking velocity by a single factor $s \in [0, 1]$ so that no monitored gap closes faster than a stopping barrier $v_{\max}(d) = \min\big(\alpha(d - d_s), \sqrt{2a(d - d_s)}\big)$. The hand therefore *decelerates straight to a stop* at the safety margin — a wall, not a force field — and tangential sliding and retraction are never impeded. Because the arm is position-servoed, the barrier is evaluated at the committed *target* pose (in a scratch `MjData`), which makes the standoff robust to approach speed. A retained *CBF-QP* alternative gives a hard forward-invariance guarantee but re-optimizes tracking, sliding the hand sideways around obstacles. A torque-domain *soft wall* extends the same barrier to the back-drivable "Soft" (compliance) controller. In closed-loop "drive the palm into the chest" tests the stop law holds a +3.8 cm standoff that is flat across operator speeds, whereas the unconstrained controller penetrates 6.5 cm to 11.5 cm; the median solve runs at 0.81 ms. Implementation: `mabel/self_collision.py`, `mabel/wbc.py`, `mabel/wbc_runtime.py`.

Keywords: Self-collision avoidance, velocity damper, control barrier functions, quadratic programming, differential inverse kinematics, capsule distance, real-time WBC.

1 Overview

The teleoperation controller (`TeleopController`, driven by keyboard, a mock stream, or a live Apple Vision Pro) maps operator wrist poses to a desired bimanual end-effector twist and resolves it through a weighted damped-least-squares (DLS) differential IK with a null-space posture task (`whole_body_control.pdf`). That solver is a single unconstrained linear-algebra step: fast, but blind to the robot's own geometry. The self-collision layer keeps the controller's free-space behaviour identical and adds a safety response near the body. It is toggleable and off by default (`config.SELF_COLLISION_ENABLE`); a single flag chooses the response law (`config.SC_METHOD`):

*Companion notes: whole-body IK (`whole_body_control.pdf`), arm control & compliance (`arm_control.pdf`), base motion (`motion_model.pdf`).

1. `"stop"` **(default)** — a velocity-scaling damper that stops the hand at the margin *along the operator's intended direction*, with no sideways redirection (§4). This is the law MABEL ships.
2. `"cbf"` — a control-barrier-function QP that re-optimizes tracking subject to a hard barrier; it guarantees invariance but slides the hand around obstacles, which the operator feels as a force field (§5).

A third response, a torque-domain *soft wall*, applies the same barrier to the back-drivable "Soft" compliance controller (§6). All three share the collision model (§2) and distance kinematics (§3).

2 Collision model: capsule bounding volumes

Each protected link i is wrapped in a capsule $\mathcal{C}_i = (s_i^0, s_i^1, r_i)$: the Minkowski sum of the core segment $\overline{s_i^0 s_i^1}$ with a ball of radius r_i. Capsules are the tightest *smooth* bounding volume for an elongated link and, unlike a box, have an everywhere-differentiable distance, which the constraint gradient below requires. We wrap the two forearms and two hands (the links actuated by the arm DOFs) and the torso and head (the obstacles they reach). Endpoints are captured once in each anchor body's frame at the rest pose, $s_i^{\{0,1\}} = R_b^\top (p_{\text{world}}^{\{0,1\}} - o_b)$, so they ride rigidly with body b and need no re-fitting as the configuration changes. Of the candidate capsule pairs we monitor only the eight that can physically meet — hands against torso, head, the opposite hand and opposite forearm, plus forearm–forearm. Pairs of *kinematically adjacent* links (parent/child within `SC_ADJ_HOPS` tree hops) are excluded automatically: adjacent links cannot self-collide, so checking them only wastes compute and risks a false constraint at rest.

Signed distance. For capsules a, b the surface gap is

$$d_{ab}(q) = \|c_a - c_b\| - r_a - r_b, \qquad (c_a, c_b) = \operatorname*{arg\,min}_{x \in \overline{s_a},\, y \in \overline{s_b}} \|x - y\|, \tag{1}$$

where (c_a, c_b) are the closest points of the two core segments, computed in closed form by the clamped segment–segment routine of Ericson [4] (a handful of dot products; no iteration). The unit separation is $n = (c_a - c_b)/\|c_a - c_b\|$. The capsule SDF (1) is exactly $\operatorname{dist}(\overline{s_a}, \overline{s_b})$ shifted by the radii, so the r_i are honest physical inflation of the links plus a comfort buffer; they are tuned (Table 1) so the rest pose clears every monitored pair by $> 6\,\text{cm}$.

3 Distance kinematics and the stopping barrier

Gradient by the envelope theorem. The witness points c_a, c_b are themselves minimizers in (1). By the envelope theorem, the configuration derivative of the optimal value equals the partial derivative at *fixed* witness points; hence, treating c_a, c_b as material points of bodies a, b,

$$\dot{d}_{ab} = n^\top (\dot{c}_a - \dot{c}_b) = n^\top \big(J_a(c_a) - J_b(c_b)\big)\, \dot{q} \;\equiv\; a_{ab}^\top \dot{q}, \tag{2}$$

where $J_a(c_a) \in \mathbb{R}^{3 \times n_v}$ is the translational point Jacobian of the material point c_a on body a (MuJoCo `mj_jac`), and $\dot{q}$ the joint velocity. The row vector $a_{ab} = n^\top (J_a - J_b)$ maps a joint velocity to the rate of change of the gap; $-\dot{d}_{ab} = -a_{ab}^\top \dot{q}$ is the *closing speed*. Equation (2) is exact to first order and needs only the two point Jacobians at the witnesses. Only pairs within an *influence distance* $d_{ab} < d_i$ are considered active; far pairs cost only the segment-distance evaluation.

The stopping barrier. We cap the admissible closing speed of every active pair by a distance-dependent ceiling

$$-\dot{d}_{ab} \;\le\; v_{\max}(d_{ab}), \qquad v_{\max}(d) = \min\Big(\underbrace{\alpha\,(d-d_s)}_{\text{CBF ramp}},\; \underbrace{\sqrt{2\,a\,(d-d_s)}}_{\text{braking distance}}\Big), \tag{3}$$

with safety margin d_s, gain α, and stopping deceleration a (`SC_STOP_DECEL`). Both terms vanish at $d = d_s$ and are positive for $d > d_s$, so the allowed approach goes to zero exactly at the margin — a head-on motion is brought to rest there. The linear term is the classical velocity damper of Faverjon and Tournassoud [1] / a zeroing control-barrier function [2]; it keeps the *final* approach gentle. The $\sqrt{2a(d-d_s)}$ term is a braking-distance limit — the fastest speed from which a body decelerating at a can still stop within the remaining gap — and dominates at speed, so a fast incoming motion is slowed early enough to stop before the margin even given the first-order error of the velocity-level model.

4 The stop law (default)

Let $\dot{q}_{\text{trk}}$ be the operator's tracking velocity — the unconstrained weighted-DLS solution with null-space posture (`whole_body_control.pdf`). The stop law applies a single *uniform* scale,

$$\dot{q} \;=\; s\,\dot{q}_{\text{trk}}, \qquad s \;=\; \min\Big(1,\; \min_{ab\,\text{active}} \frac{v_{\max}(d_{ab})}{\max(0,\; -a_{ab}^{\top}\dot{q}_{\text{trk}})}\Big), \tag{4}$$

where the inner minimum runs over active pairs whose tracking motion is *closing* ($-a_{ab}^{\top}\dot{q}_{\text{trk}} > 0$); a pair that is stationary or separating imposes no limit. Three properties follow directly and are exactly what teleoperation wants:

- **Zero deflection.** $\dot{q} = s\,\dot{q}_{\text{trk}}$ is collinear with $\dot{q}_{\text{trk}}$, so the end-effector path is unchanged — the hand decelerates *straight along the operator's intended direction* and stops. There is no sideways push: the operator never fights a "force field."
- **Tangential / retreat motion is free.** A motion that slides along an obstacle or pulls away has $-a_{ab}^{\top}\dot{q}_{\text{trk}} \le 0$ for every pair, so $s = 1$ and the command passes through untouched.
- **Hard stop at the margin.** As $d_{ab} \downarrow d_s$, $v_{\max} \to 0$, so any residual closing motion is scaled to zero: the hand arrives at the margin with zero approach speed.

Why evaluate at the target pose. The arm is driven by stiff *position* servos: `solve` integrates $\dot{q}$ into joint targets q^{tgt} and commands them, and the physical configuration q *lags* q^{tgt}. If the barrier (4) is evaluated at the lagging physical pose, the target can be driven past the margin while the arm is still catching up, and the arm then settles inside the margin (penetration). We therefore evaluate d_{ab}, n, and a_{ab} at the committed *target* pose — the configuration the servos are driving toward — so the target itself never crosses the margin and the physical arm settles outside it. The target pose is realised in a scratch `MjData` (`mj_kinematics` + `mj_comPos` only), leaving the live simulation untouched; the cost is one extra forward-kinematics evaluation per tick, paid only when the guard is on. Empirically this makes the standoff essentially independent of approach speed (Fig. 1), unlike a physical-pose gate.

Equivalence in free space. With no pair closing too fast, $s = 1$ and the output is exactly the weighted-DLS tracking solution of `whole_body_control.pdf`: the stop law is behaviour-preserving away from the body and engages only inside the influence band.

5 The CBF-QP alternative (`SC_METHOD="cbf"`)

For applications that need a formal forward-invariance guarantee the layer retains the control-barrier-function QP. Substituting (2) into the linear CBF condition $\dot{d}_{ab} \geq -\alpha(d_{ab} - d_s)$ gives one linear inequality per active pair, $-a_{ab}^{\top}\dot{q} \leq \alpha(d_{ab} - d_s)$, i.e. $G\dot{q} \leq h$. The IK step becomes the strictly-convex QP

$$\min_{\dot{q}} \ \tfrac{1}{2}\|J\dot{q} - v\|^2 + \tfrac{1}{2}\lambda^2 \dot{q}^{\top} W \dot{q} + \tfrac{1}{2}\mu\|\dot{q} - \dot{q}_{\text{post}}\|^2 \quad \text{s.t.} \quad G\dot{q} \leq h, \ \ \dot{q}_{\text{lo}} \leq \dot{q} \leq \dot{q}_{\text{hi}}, \tag{5}$$

with J the stacked bimanual task Jacobian, v the desired task twist, W the per-DOF DLS weight, λ the damping, μ a small posture weight, and the box a per-DOF velocity/joint-limit damper. With no active inequality its minimizer is exactly the weighted DLS, so it too is free-space-preserving. It is solved by the Goldfarb–Idnani dual active-set method [3] (`quadprog`), falling back to OSQP [5] and then to a box-clamped solve. **Trade-off:** because (5) *re-optimizes tracking* subject to the barrier, when a pair binds it trades the blocked normal motion for tangential motion — the hand slides *around* the obstacle rather than stopping in front of it. This sideways redirection is what the operator experiences as a gravity/force field fighting the intended direction, and is the reason the stop law is the default.

6 Compliance soft wall (torque domain)

MABEL's "Soft" controller (`WBCEngine`, see `arm_control.pdf`) is back-drivable: the arm joints are gravity-compensated and the operator pushes them directly, so there is no commanded velocity to scale. The same barrier instead becomes a generalized *braking force*. For each active pair, with closing rate $\dot{d}_{ab} = a_{ab}^{\top}\dot{q}$ and allowed rate $\dot{d}_{\min} = -\alpha(d_{ab} - d_s)$,

$$f_{ab} = k_v \ \max(0, \ \dot{d}_{\min} - \dot{d}_{ab}) \ + \ k_p \ \max(0, \ d_s - d_{ab}) \ \geq 0, \qquad \tau \mathrel{+}= a_{ab}^{\top} f_{ab}, \tag{6}$$

added to the joint efforts. The k_v term brakes only motion that is closing faster than allowed; the k_p term springs the link back out once it is inside the margin. Both act *along the separation direction* $a_{ab}^{\top}$ and are zero outside the influence band, so free dragging is unaffected and a dragged hand decelerates to a soft stop at the margin instead of being pushed through the body.

7 Implementation and latency engineering

The hot path is `WholeBodyIK.solve` in `mabel/wbc.py`; the collision model and both barriers live in `mabel/self_collision.py`, and the compliance soft wall is wired in `mabel/wbc_runtime.py`. Low latency comes from removing Python and allocation overhead, not from approximation:

- **Cached indices.** Joint `qpos` addresses and limit ranges are gathered once at construction; the per-tick state read is a single vectorised `data.qpos[adr]` rather than n named lookups.
- **Preallocated buffers.** Task- and point-Jacobian scratch and the stop law's target-pose `MjData` are allocated once and overwritten in place.
- **Broad phase by influence.** A pair pays for its two `mj_jac` calls only when its cheap segment distance falls below d_i.
- **GC.** For deterministic timing the real-time loop runs with the cyclic garbage collector disabled; the figures below are measured this way.

Table 1: Measured solve latency (6000 samples, GC off) and parameters.

controller	mean [ms]	p50 [ms]	p99 [ms]	added vs DLS
bare DLS (guard off)	0.40	0.33	1.51	—
STOP (default)	0.92	0.81	2.56	+0.48 ms
CBF-QP	0.98	0.84	3.45	+0.51 ms

parameters (`config.SC_*`)						
d_i infl.	d_s margin	α	a decel	k_v/k_p (wall)	box cap / limit	radii (fore/hand/torso/head)
6 cm	2.5 cm	6 s	2 m/s^2	40 / 600	4 rad/s / 8 s	35/55/90/120 mm

8 Experiments

All results are from `scripts/tools/self_collision_study.py` on the `mabel_sim` scene, arms-only IK ($n = 16$), MuJoCo 3.5, $\Delta t = 2$ ms. The "gap" is the tightest surface distance over all monitored capsule pairs; negative means the inflated capsules overlap (the conservative radii put the true links still further apart).

Intrusion (Fig. 1). The right palm is commanded straight at the torso centre (command lerp 0.004/tick). Unguarded, the hand drives 6.5 cm of capsule penetration into the chest. The stop law decelerates the hand straight along its path and holds it +3.8 cm clear; the CBF-QP holds +1.5 cm (and slides the hand sideways to do so). Neither contacts the body.

Speed robustness (Fig. 1). Sweeping the operator command speed (target lerp 0.002–0.020/tick) exposes the key difference. The stop law, gated at the *target* pose, holds an almost constant standoff (+3.86 cm down to +3.55 cm) across the whole range. The CBF-QP, gated at the physical pose, tightens monotonically from +2.05 cm to +0.03 cm (nearly grazing) as the approach quickens, while the unguarded controller penetrates 8 cm to 11.5 cm.

Barrier envelope (Fig. 2). Plotting the realised approach speed $-\dot{d}$ against the gap during a stop-law run shows every sample sitting on or below the ceiling $v_{\max}(d)$ of (3): the braking-distance branch binds at larger gaps and the linear branch near the margin, and the allowed approach $\to 0$ as $d \to d_s$.

Compliance soft wall (Fig. 2). Dragging the right palm into the chest with a constant 40 N pull under the back-drivable compliance law: with the soft wall off the hand is dragged 2.7 cm into the chest capsule; with it on the hand stops +0.4 cm clear and holds there against the pull.

Latency (Fig. 3, Table 1). Over 6000 solves with a constraint active about half the time, the stop law runs at a median of 0.81 ms (mean 0.92 ms), versus 0.33 ms for the bare DLS — the extra cost is the one target-pose forward-kinematics evaluation. The CBF-QP is comparable at the median (0.84 ms) but has a heavier tail. Both medians sit well inside the 2 ms physics tick; occasional tail spikes (p99 2.6 ms to 3.4 ms; max 10 ms to 20 ms) are OS/allocator jitter, not algorithmic, and are absorbed by the bridge's per-frame substepping. The guard is off by default, so the nominal teleop path pays none of this.

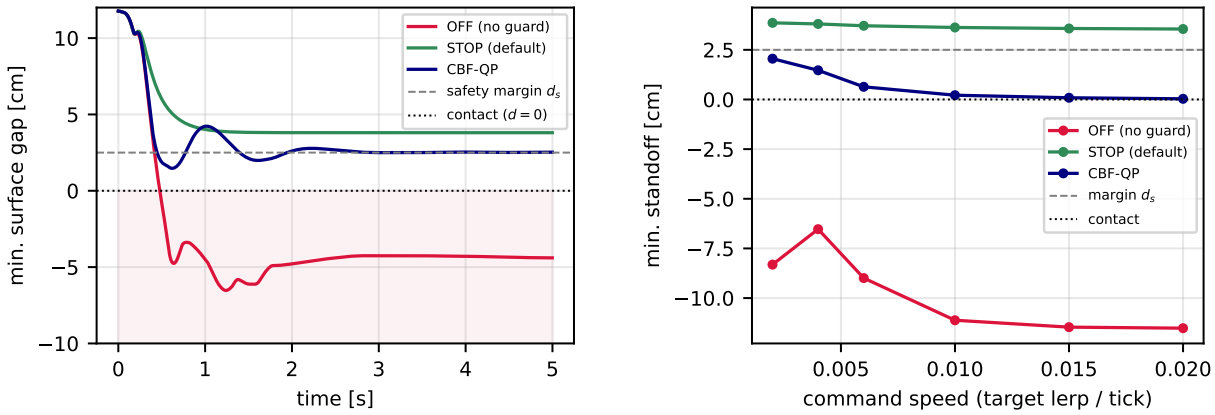


Figure 1: Left: tightest surface gap while the palm is driven at the chest — OFF penetrates 6.5 cm, STOP holds +3.8 cm, CBF holds +1.5 cm. Right: minimum standoff vs operator command speed — STOP is flat (target-pose gating), CBF tightens to near contact, OFF penetrates throughout.

9 Usage

The guard is *off by default* (`config.SELF_COLLISION_ENABLE = False`); the response law is `config.SC_METHOD` (`"stop"` default, `"cbf"` alternative). It is toggled at runtime via `TeleopController.set_self_collision(on, method=None)` / `toggle_self_collision()` (mirrored on `WBCEngine` for the compliance soft wall); the live state appears in `TeleopController.status()["self_collision"]`.

```
mjpython scripts/teleop.py                  # plain teleop; press 'o' to toggle
mjpython scripts/teleop.py --self-collision     # start with avoidance ON
mjpython scripts/compliance.py --region arm --self-collision   # soft wall ON
python   scripts/tools/self_collision_study.py  # regenerate these figures
```

On the live bridge the default is off; enable it with `sim_teleop_bridge.py --self-collision` or `MABEL_SELF_COLLISION=1`. Tunables live in `mabel/config.py` (`SC_*`): the influence and margin distances, gain α, stopping deceleration, soft-wall gains k_v, k_p, the CBF-QP box gains, and the capsule radii.

10 Limitations

- **Conservative standoff vs. tight margin.** The stop law's target-pose gating keeps the *target* at the margin, so the physical arm (which lags the target) settles further out — a $\sim$1.3 cm buffer above d_s here. This errs safe but reaches slightly less far than the CBF-QP, which gates on the physical pose and grazes the margin (at the cost of speed-dependent tightening and sideways deflection).
- **First-order, velocity-level model.** Both barriers use the first-order $\dot{d} = a^\top \dot{q}$; the braking-distance term and target-pose gating absorb the discretisation error in the stop law, but a sufficiently violent command can still dip toward the margin before stopping. The conservative capsule radii keep this off the true geometry.
- **Modelled geometry only.** Capsules approximate the true links; the fingers are folded into the hand capsule, and the low-value forearm–torso and forearm–head pairs are omitted (the forearms

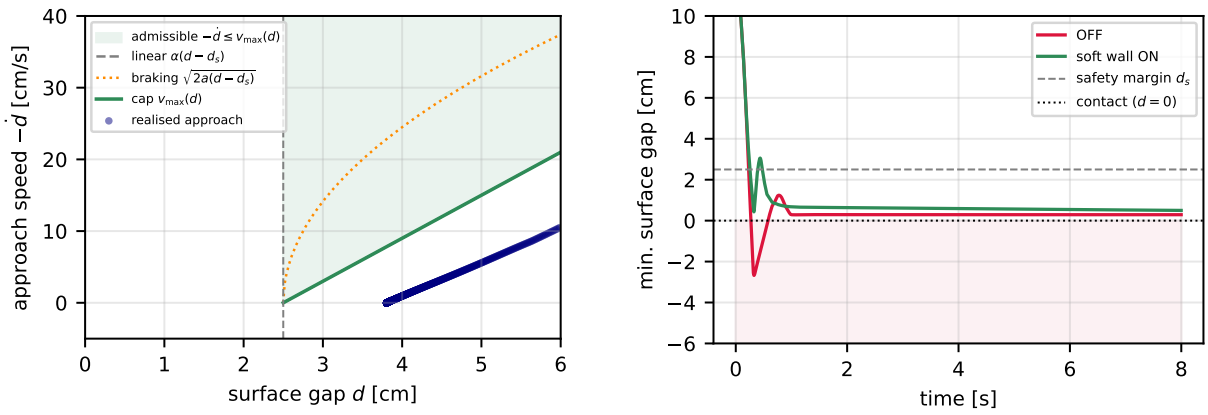


Figure 2: Left: the stop barrier $v_{\max}(d) = \min(\alpha(d-d_s), \sqrt{2a(d-d_s)})$ and the realised approach, which stays inside it and vanishes at d_s. Right: compliance soft wall — a 40 N drag into the chest penetrates 2.7 cm with the wall off and is held +0.4 cm clear with it on.

rest close to the torso and the leading hand is guarded there instead). Extend `SC_RADII` and the pair list in `self_collision.py` for finer coverage at a small latency cost.

- **CBF-QP feasibility.** With many simultaneously active constraints the QP can in principle be infeasible; the box limits and a single dominant collision row keep this from occurring in practice, and the solver falls back rather than failing. The stop law cannot be infeasible — it only ever scales the command down.

References

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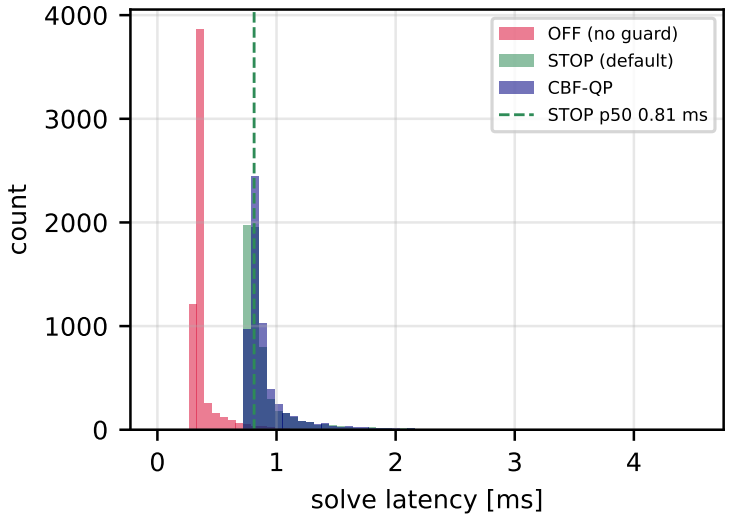


Figure 3: Per-solve latency (GC disabled): bare DLS vs the STOP law vs the CBF-QP. The STOP median is 0.81 ms, comfortably inside the 2 ms physics budget; tail spikes are OS/allocator jitter.